SoftwareONE

EMPOWERING COMPANIES TO TRANSFORM

STC health

ASSESSMENT RESULTS & MIGRATION PLAN

softwareONE

AGENDA

01 Assessment Summary & Results

02 Recommended path forward

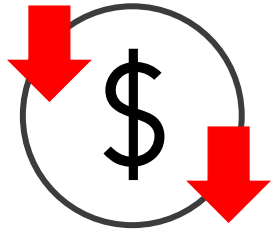
03 Commercials & Next Steps

01

ASSESSMENT SUMMARY & RESULTS

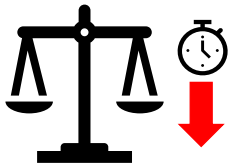
Our Understanding

STC is transitioning to a SaaS business model and about 50% complete. Planning to migrate and/or modernize the legacy Immunization Registry systems as part of this transition. Looking for best path forward while achieving the following benefits .



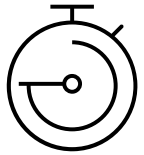
Decrease overall Technical Cost

Reduce Database related incidents



Reduce Time to scale

Increase self service of envs



Reduce time to resolve

Assessment Summary

SoftwareONE analyzed and assessed code samples to understand the complexity, dependencies and find best path forward to meet STC's objectives. We collaborated with members of Leadership, DevOps, Application and DBA teams to understand dependencies and potential paths forward.

How we did our assessment

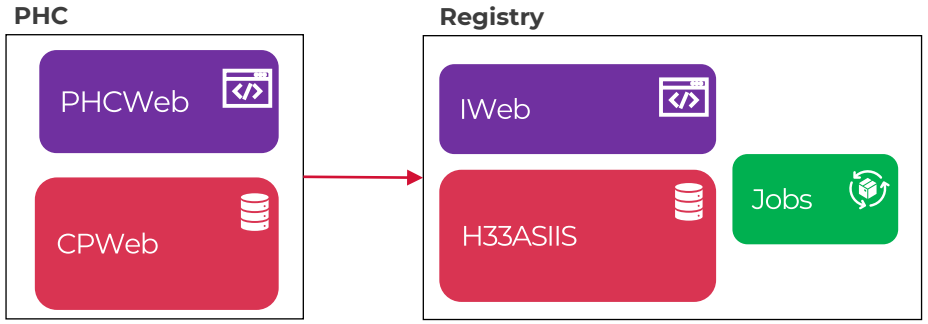
AWS Schema Conversion Tool

ora2pg

CAST HIGHLIGHT
Code-level portfolio analysis. FAST.

Custom Scripts & analysis

Scope of Assessment



Scope details

1 Instance	192k LOC	589 Tables	54 Functions
70 Triggers	13 Views	3 Java Classes	8 Packages

30+ TB Database Size

Oracle 12c Release 1

Assessment Summary

In order to migrate, about 42% of all ***code*** objects need to be **converted manually**. SPs, Functions & Jobs contain the bulk of complexity. The ***storage*** objects, such as Table, Constraints, Sequences can be mostly converted with automated tools.

AWS SCT Report

Category	# of objects	# automatically converted	# with simple actions		# with medium-complexity actions		# with complex actions		Total lines of code	% needing Manual conversion
JOB	5	0	0	0%	0	0%	5	100%	88	100%
PROCEDURE	161	10	26	16%	16	10%	109	68%	127626	94%
FUNCTION	54	14	24	44%	2	4%	14	26%	21539	74%
TRIGGER	70	27	38	54%	1	1%	4	6%	7882	61%
PACKAGE_PROCEDURE	38	23	2	5%	1	3%	12	32%		39%
PACKAGE_FUNCTION	22	20	1	5%	0	0%	1	5%		9%
TABLE	582	566	8	1%	0	0%	8	1%	20956	3%
PACKAGE_COLL_TYPE	2	2	0	0%	0	0%	0	0%		0%
PACKAGE_CONSTANT	5	5	0	0%	0	0%	0	0%		0%
PACKAGE_VARIABLE	4	4	0	0%	0	0%	0	0%		0%
SEQUENCE	318	318	0	0%	0	0%	0	0%	1906	0%
PACKAGE_TYPE	6	6	0	0%	0	0%	0	0%		0%
PACKAGE	8	8	0	0%	0	0%	0	0%	3969	0%
MATERIALIZED_VIEW	7	7	0	0%	0	0%	0	0%	815	0%
VIEW	13	13	0	0%	0	0%	0	0%	437	0%
INDEX	576	576	0	0%	0	0%	0	0%	6336	0%
CONSTRAINT	629	629	0	0%	0	0%	0	0%	425	0%
SCHEMA	1	1	0	0%	0	0%	0	0%	1	0%

94% Manual of which 68% are Complex

Assessment Summary

Manual conversion effort is made up of over 3000 actions and about half aren't simple. The issues have high levels of concentration and so highly repeatable. For ex: Txn flow & control make up bulk of issues with high complexity.

Manual Action Complexity breakdown	Count	Percent
⊞ Complex	1412	42%
⊞ Medium	242	7%
⊞ Simple	1713	51%
Grand Total	3367	100%

⊟ Simple			1713	51%
Converted functions depends on the time zone settings			1317	36%
Unable to resolve objects			162	8%
Dynamic SQL parameter dependency			86	2%
Unable to convert statements with dynamic SQL statement			82	2%
PostgreSQL doesn't support SAVEPOINT, ROLLBACK TO SAVEPOINT inside of routines			28	1%
This statement references to a synonym			12	0%
Table contains LOB column(s)			11	0%
PostgreSQL doesn't support a referen			5	0%
PostgreSQL error code type is not a nu			5	0%
Unable to convert hints			2	0%
Converted code execution results mig			1	0%
Converted code might be incorrect			1	0%
PostgreSQL expands fractional seconc				
	⊟ Medium	242	7%	
	Unable to automatically transform the SELECT statement	85	2%	
	Converted code might be incorrect because of the dynamic SQL	49	1%	
	PostgreSQL doesn't have a data type ROWID	36	1%	
	PostgreSQL doesn't support a ROWID	23	1%	
	Unable to transform statement due to references to unresolve			
	Encrypted Objects			
	Unsupported DDL command			
	PostgreSQL routines cannot have more than 100 IN-INOUT argu			
	Unable to convert system object			
	PostgreSQL doesn't support the UPDATE statement for a subqu			
	PostgreSQL doesn't support TYPE ... IS REF CURSOR usage			

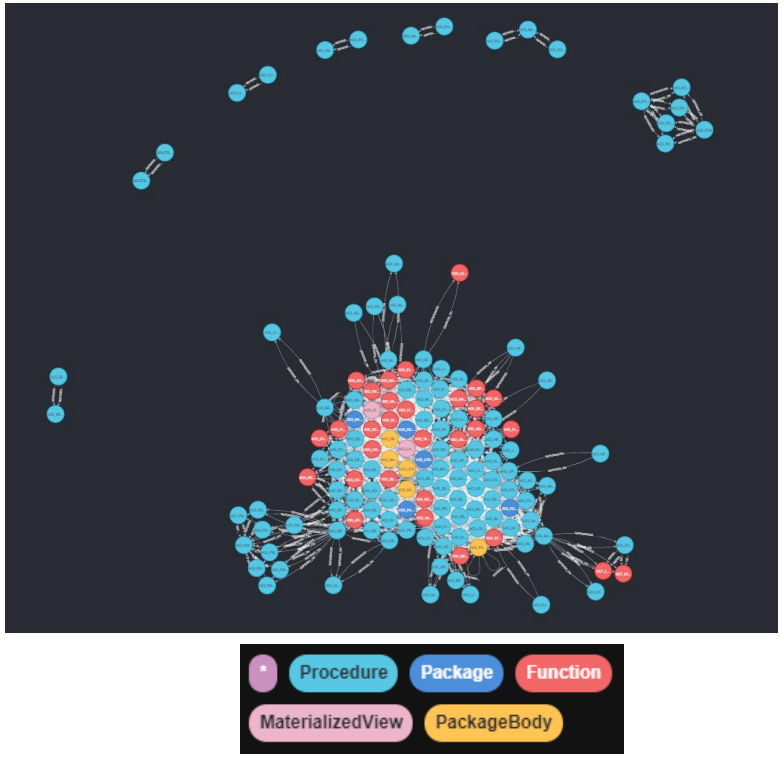
⊟ Complex	1412	42%
Transaction control is only possible from the top level or nested call invocations without any other intervening command	622	15%
A transaction cannot be ended inside a block with exception handlers	428	11%
Internal Converter error occurred	277	14%
Unable to convert functions	20	1%
PL/SQL pragma is unsupported	16	0%
Dynamic SQL table dependency	11	0%
Unable automatically convert statements that explicitly apply or cancel a transaction	9	0%
Unable to convert the UPDATE statement with multiple-column subquery in SET clause	7	0%
Encrypted Objects	6	0%
Unable to convert objects	5	0%
Resultset could vary from Oracle	4	0%
PostgreSQL doesn't support conditional predicates	2	0%
Unable to transform statement due to references to unresolved object	1	0%
Unable to automatically transform the SELECT statement	1	0%
Unable to convert Java stored routine	1	0%
Unparsed SQL	1	0%
PostgreSQL routines cannot have more than 100 IN-INOUT arguments	1	0%

Assessment Summary

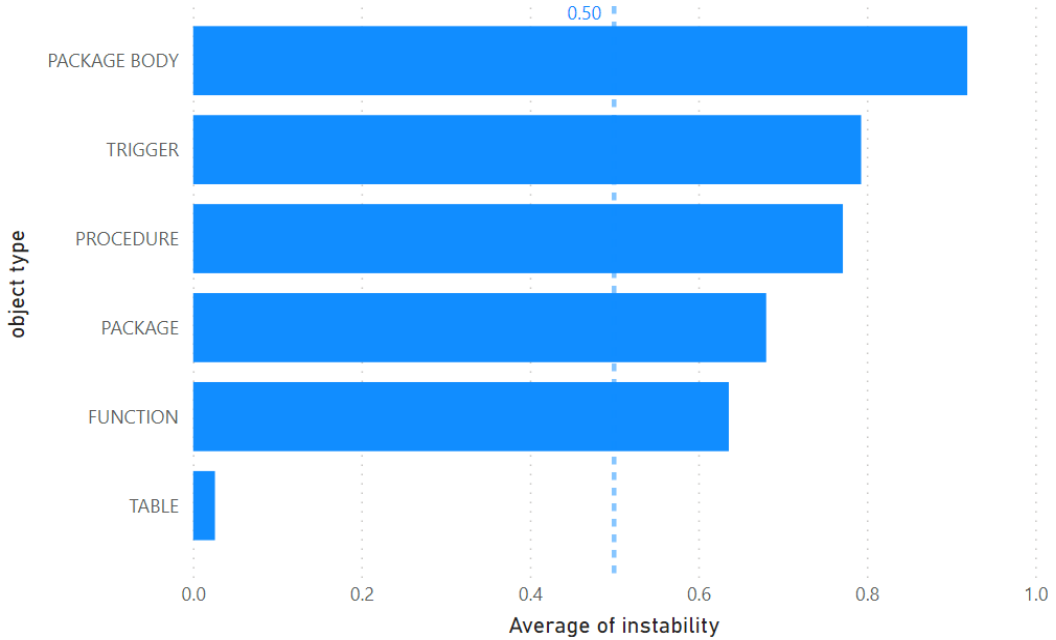
The code objects are highly coupled and cannot be migrated incrementally. This makes the migration process risky as there is potential for rework from conflict with other ongoing workstreams in STC.

For ex: a schema refactoring in Oracle could impact the migration process.

Coupling of objects



Degree of coupling and resulting instability



02

RECOMMENDED PATH FORWARD

Assessment Recommendations

Migrating from Oracle to AWS RDS Aurora can help achieve some of the desired outcomes such as cost reduction and improved operational readiness.

However, given the amount of technical debt and fragility of the code units, full set of desired outcomes cannot be achieved with migration alone. Therefore, we recommend starting the modernization process in parallel.

- Prove out migration tech approach using a short POC.
- Reduce performance issues by optimizing db queries and views.
- Extract domain events from the app (not CDC) to start the modernization.

Recommended Path

We estimate starting cutover of your first customer around month 12. Based on our learnings, we would plan out the migration of the remaining 15 customers in subsequent waves.

Activities	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Month 8	Month 9	Month 10	Month 11	Month 12	Month 13	Month 14	Month 15	Month 16	Month 17	Month 18	Month 19	Month 20	Month 21	Month 22	Month 23	Month 24
Migration																								
POC																								
DB Storage Objects Conversion																								
DB Code Object Conversion																								
Data conversion																								
App Query Conversion																								
Data replication																								
Configuration & Migration Env Setup																								
Migrate & Validation																								
Cutover																								
Wave 1 - I XL Customer																								
Wave 2 - TBD																								
Wave 3 - TBD (beyond month 24)																								
Modernization																								
First increment of value																								

POC Scope

Activities

- Validate estimates & complexity of manual actions.
- Create reusable solutions for top set of issues
- Dry run migration tasks for select objects:
  - *7 Procedures,*
  - *1 Function,*
  - *1 Table,*
  - *~40 Transaction control actions &*
  - *~14k LOC.*
- Validation of Data transfer & database performance
- Observability & Monitoring approach
- Flush out migration plan details

Deliverables

- Reusable technical solutions for the top manual actions
- Data validation approach
- Detailed migration plan with updated estimates

POC Plan

We will run targeted migration activities to validate estimates and feasibility of the manual conversion work. We will also spend some time to assess the first customer's (the "early adopter") path to cutover.

Activities	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Kickoff								
Migration								
SCT migration execution & validation								
Manual action execution & validation								
Metadata validation (src vs target)								
Early Adopter Analysis								
DB Performance & Health check								
Data transfer execution & analysis								
DMS behavior analysis								
PoC Closing								
Findings & Analysis								
Define approach & next steps								
Migration plan								

Modernization Scope

Our goal is to deliver the first increment of functionality that will:

1) flush out the data sync strategy to ensure consistency, idempotency and traceability.
2) drive out a repeatable approach to incremental strangulation of the legacy
3) deliver business value
4) establish meaningful feedback loops with product and end customers

Activities

- Event storming and bounded context workshops to drive out future state data domains & contracts.
- Create domain event capture approach with event level traceability & versioning (not CDC tool based)
- Create core domain data, business capability and UI services as needed.

Deliverables

- Working software – domain, capability and UI services for selected feature.
- Repeatable modernization pattern
- Prioritized roadmap for completing modernization

Modernization Plan

Activities	Sprint 1	Sprint 2	Sprint 3	Sprint 4	Sprint 5	Sprint 6	Sprint 7	Sprint 8	Sprint 9	Sprint 10	Sprint 11	Sprint 12
Inception												
Identify candidate for first increment												
Event storming & Bounded context												
Data sync design												
Scope & delivery plan												
Kickoff (Setup LZ, pipelines, provisioning)												
Data sync												
Capture domain events												
Establish contract testing & ingestion services (legacy to new)												
Event domain versioning in legacy												
Validate idempotency and consistency of sync												
Capability & UI Services												
Modernization roadmap - iWeb & PHC												

Repeatable modernization pattern

03

COMMERCIALS & NEXT STEPS

Commercials & Next Steps

The cost for the POC, Migration & Modernization – Domain extraction is **$2,004,704**. AWS may offset some of this cost contingent on meeting requirements of the AWS MAP Program.

Description	Cost
POC	$200,016
Migration	$1,438,880
Modernization – Domain Events extraction	$365,808
Estimated Cost	$2,004,704*

* Final cost subject to change pending internal review and validation

Assumptions

- Remote engagement based out of Colombia & North America.
- Team will undergo HIPAA training

Next Steps

- Lejo to send proposal & Assess phase deliverables by 10/22.
- STC to complete AWS deliverable acceptance form by 10/29.
- SWO to conduct project closure call with STC week of Nov1 and send project closure form for STC's sign-off.
- Lejo to setup call between STC, SWO & AWS to align on funding requirements (targeting Week of Nov 1).

THANK YOU!

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Get in Touch

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APPENDIX

01

ASSESSMENT DETAILS

Assessment Results

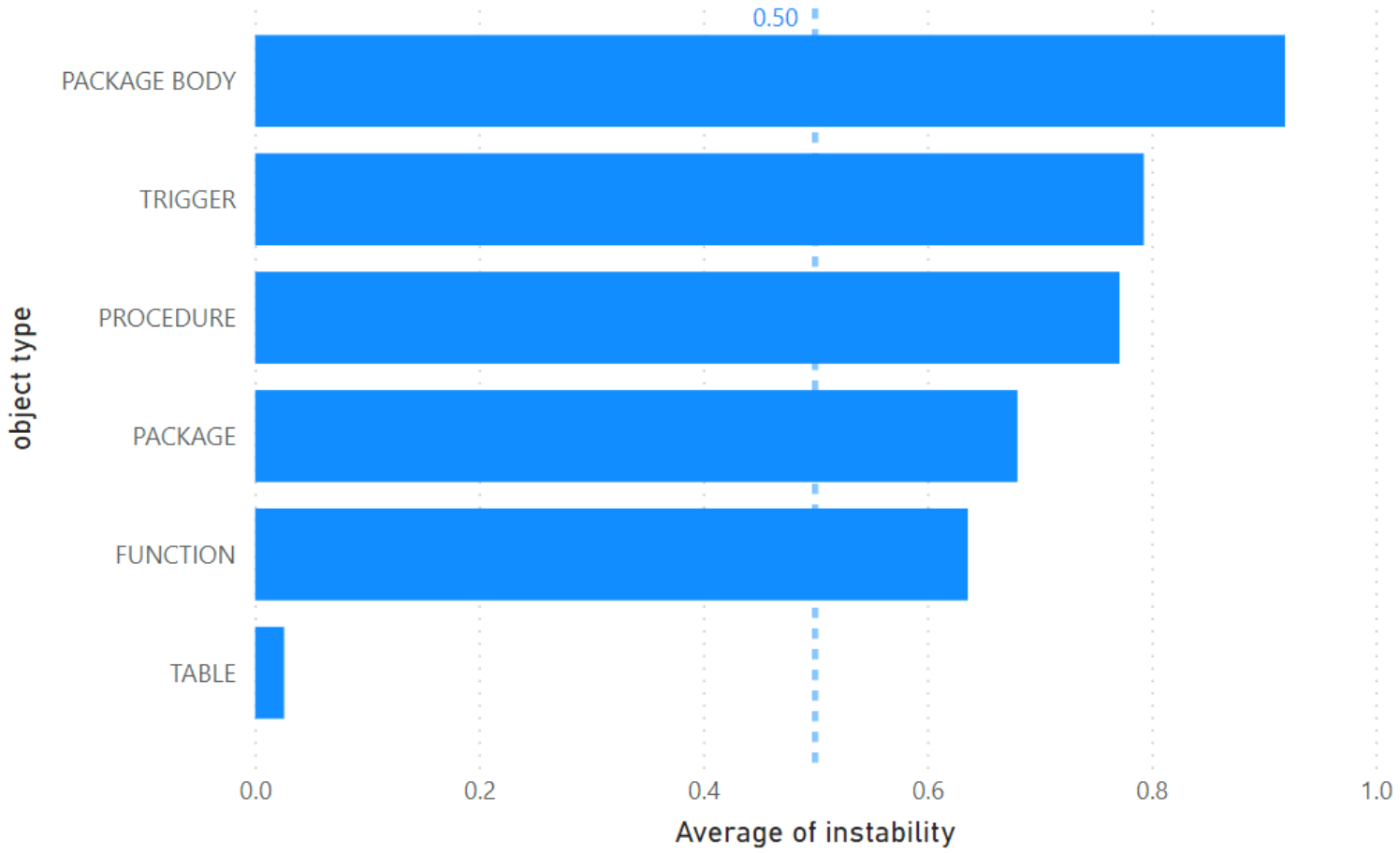
*"**Instability** is a metric used to measure the relative susceptibility of a component to breaking changes. Instability indicates that a software class, package, **subsystem** or other given module will be significantly impacted by changes elsewhere. Instability is calculated according to the following formula, where **I** is denoted by instability:*

I = Ce / (Ce + Ca)

*In the equation, **Ce** represents outgoing coupling, and **Ca** represents incoming coupling. This formula will produce a value for I that ranges from 0 to 1. Components are considered to be generally stable when this calculation places the value of **I** closer to 0. Components with tighter coupling will usually produce a value closer to 1, which indicates that it is highly susceptible to breaking changes."*

Assessment Results

Coupling

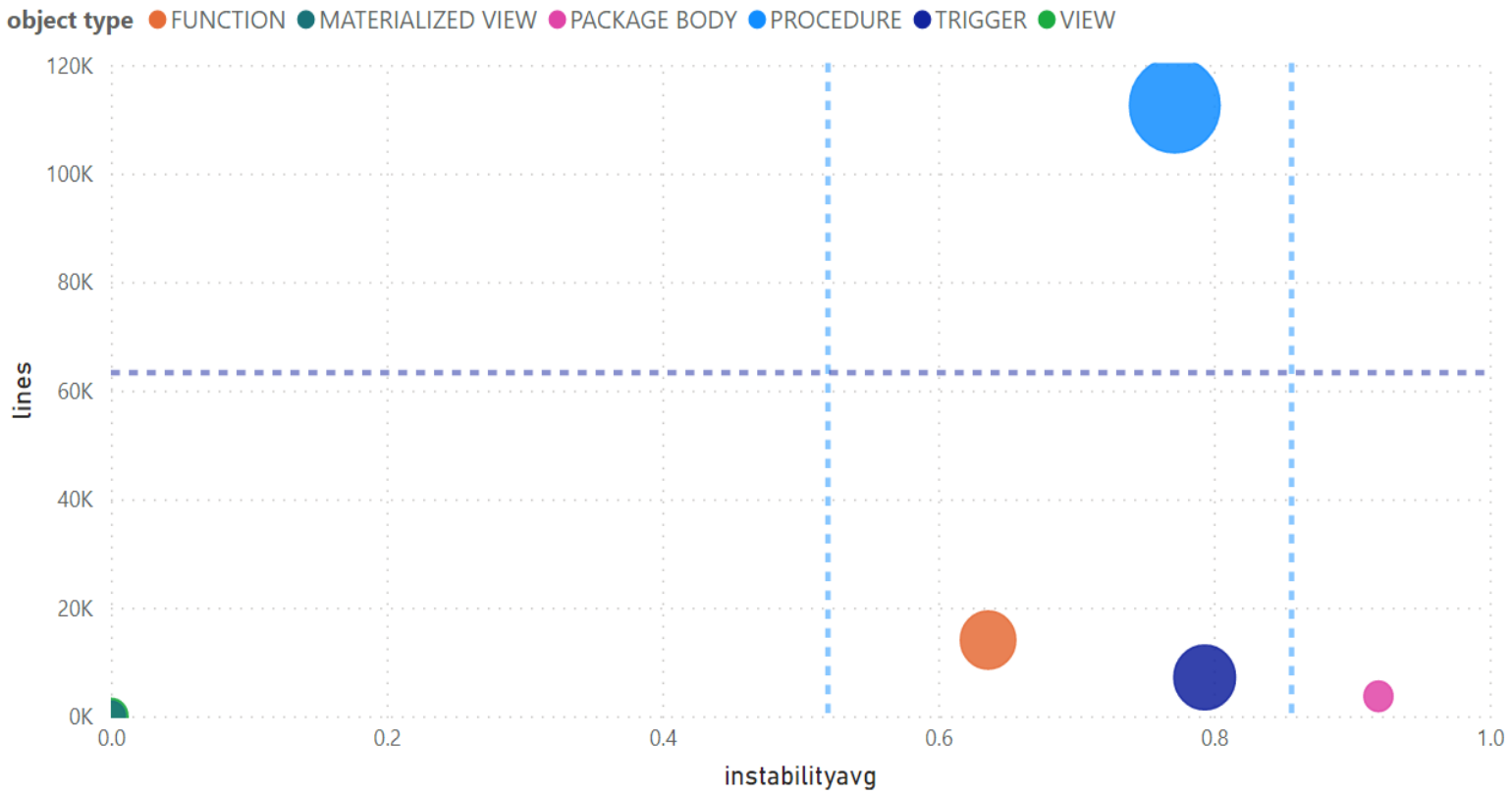


Instability vs Object Type

Components are considered to be generally stable when this calculation places the value of I ***closer to 0.***

Assessment Results

Coupling

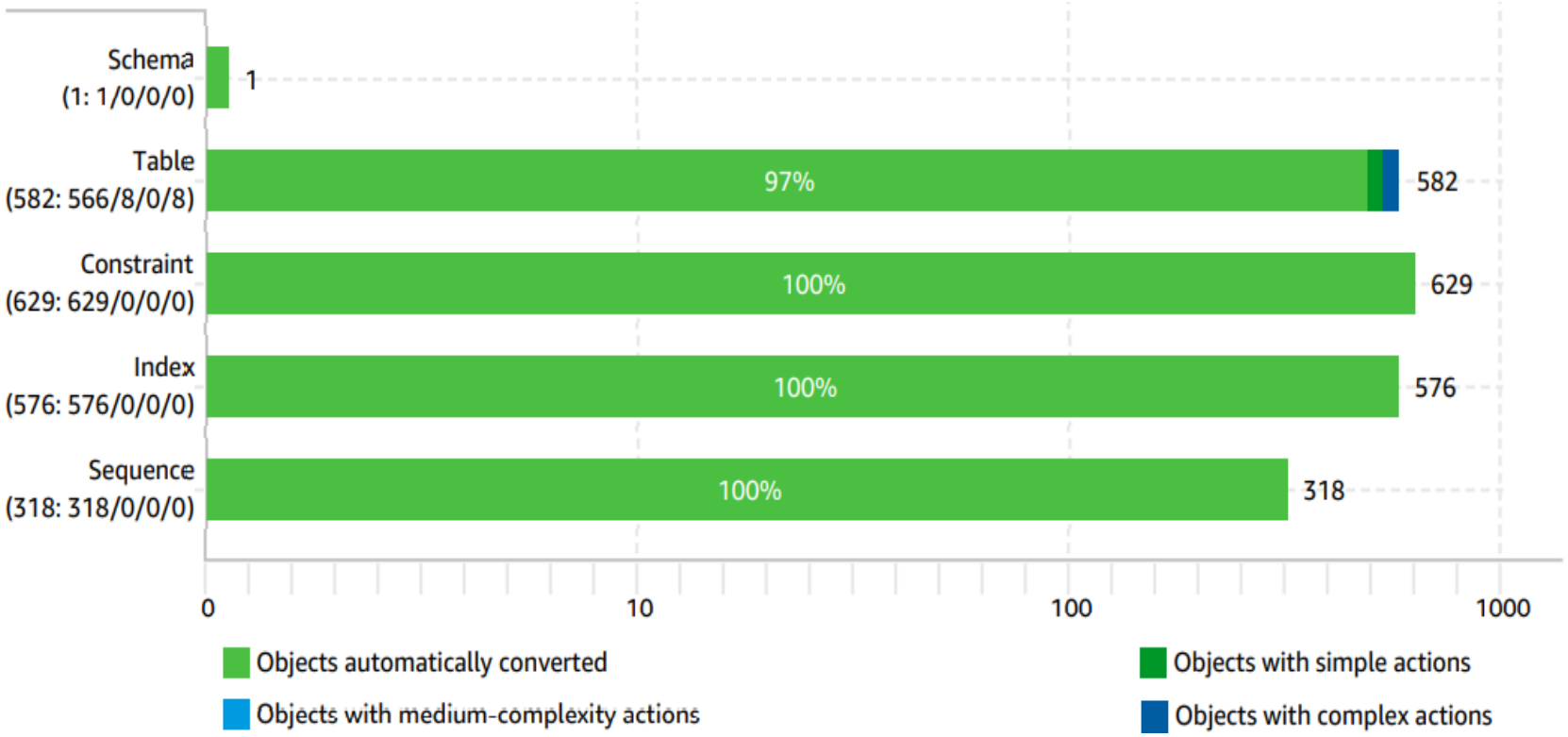


Instability vs LoC

Components are considered to be generally stable when this calculation places the value of I ***closer to 0.***

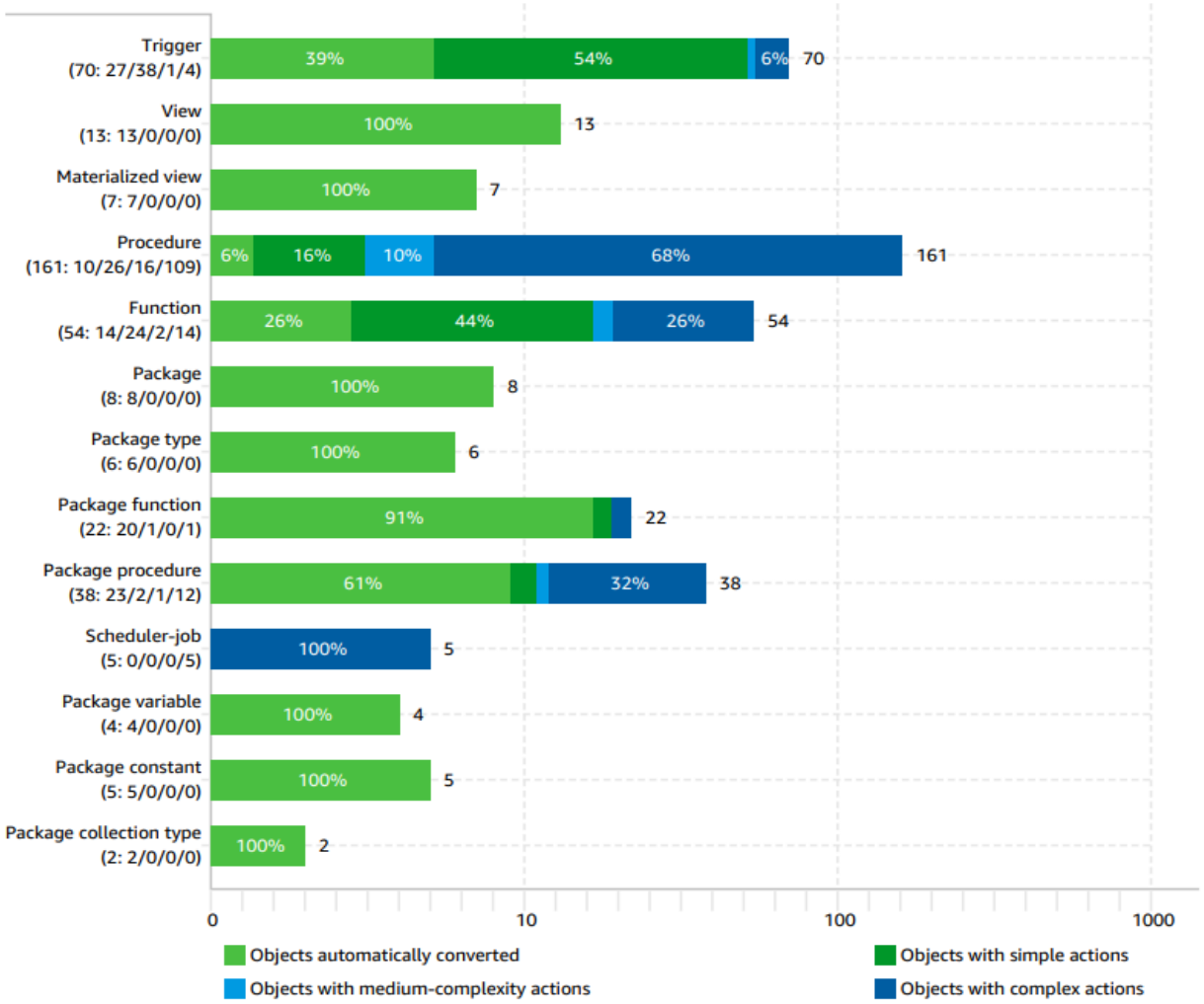
Assessment Results

Conversion statistics for database storage objects



Assessment Results

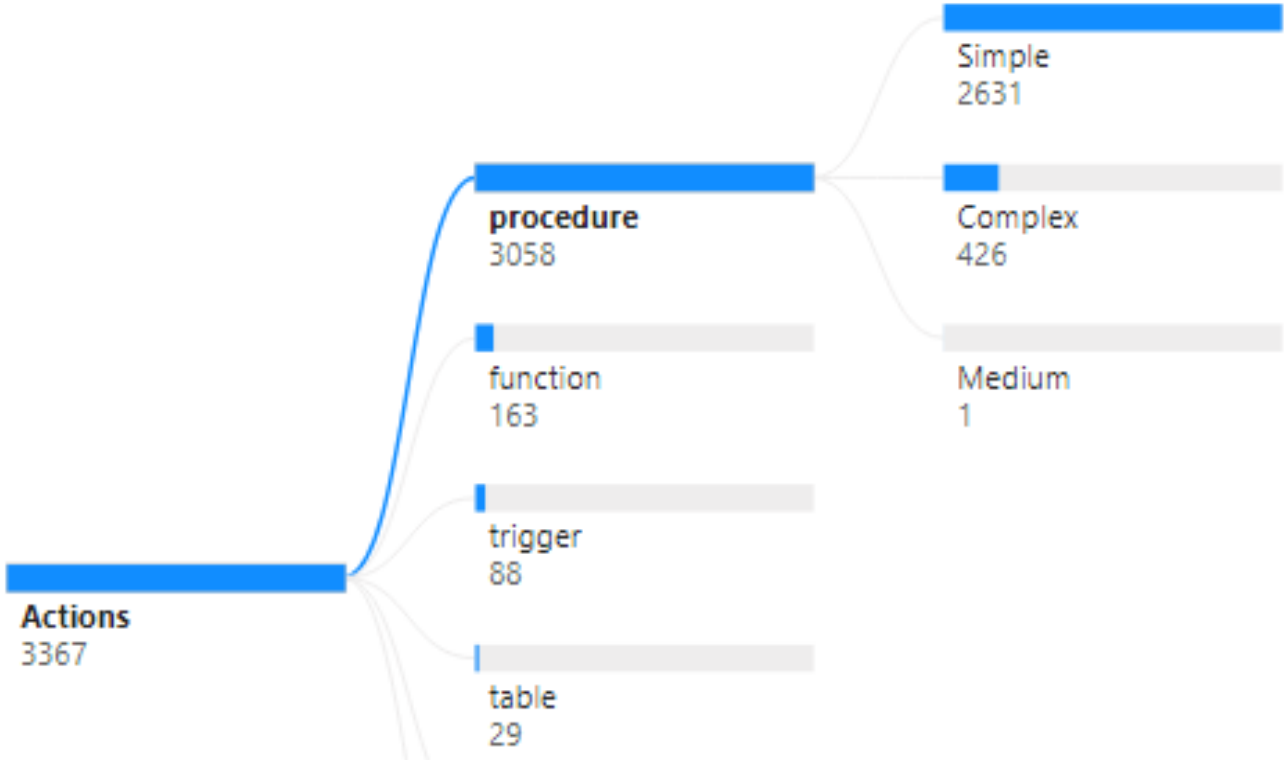
Conversion statistics for database code objects



94% *of the procedures must be manually intervened in order to execute a migration effort from Oracle to Postgres.* ***109*** *out of 161 procedures require complex manual interventions. 6% of the procedures can be converted without manual intervention.*

Assessment Results

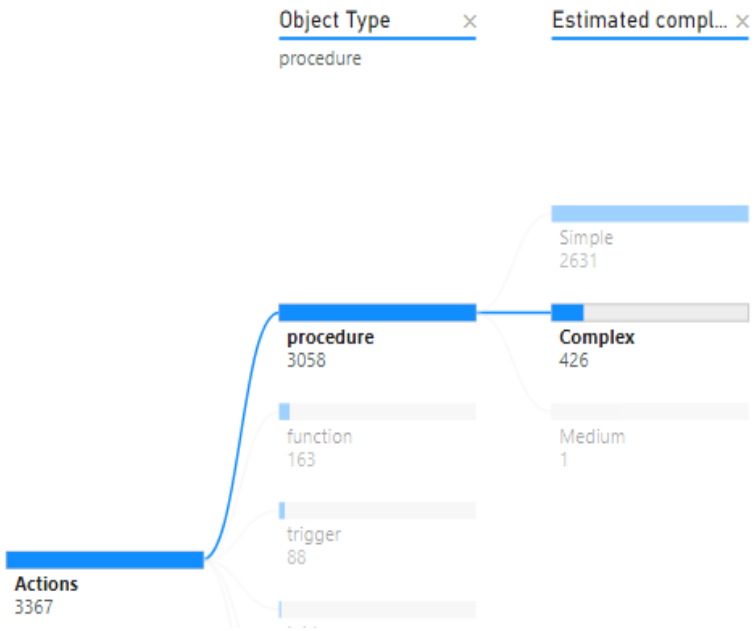
Actions required for procedures



*For code objects, **3058** actions on procedures, 163 on functions, 88 on triggers, and 24 on jobs are recommended.*

Assessment Results

Complex actions required for procedures



Occurrence	Actions
DELETELOGS	139
H33_FORECAST_VACCINATIONS	34
H33_EXPORT_EBC	22
H33_FORECAST_PERI_HEPB	15
H33_FORECAST_DTAP_GROUP	9
H33_LOT_COUNT	9
H33_DEACTIVATE_IRMS	8
H33_POP_HP_CODE	8
H33_SEPARATE	8
H33_BYPASS	5
H33_DEDUP_VACCINATIONS_BATCH	5
H33_INIT_PON_REPORT_TABLES	5
H33_MOVE_ADOPT_PT_VACCS	5
H33_PROCESS_AMBIGUOUS_PATIENT	5
H33_COUNT_ALL_ROTATEQ	4
H33_DELETE_PATIENT	4
H33_FULL_FORECAST	4

*DELETELOGS object requires **139** complex actions out of a total of 426 required for the entire set of procedures, and represents 32% of the required actions.*

Assessment Results

Estimated migration effort

SWO Estimate (P)	SCT Estimate (M)	ORA2PG Estimate (O)
6254 Total Hours	**4440** Total Hours	**3416** Total Hours

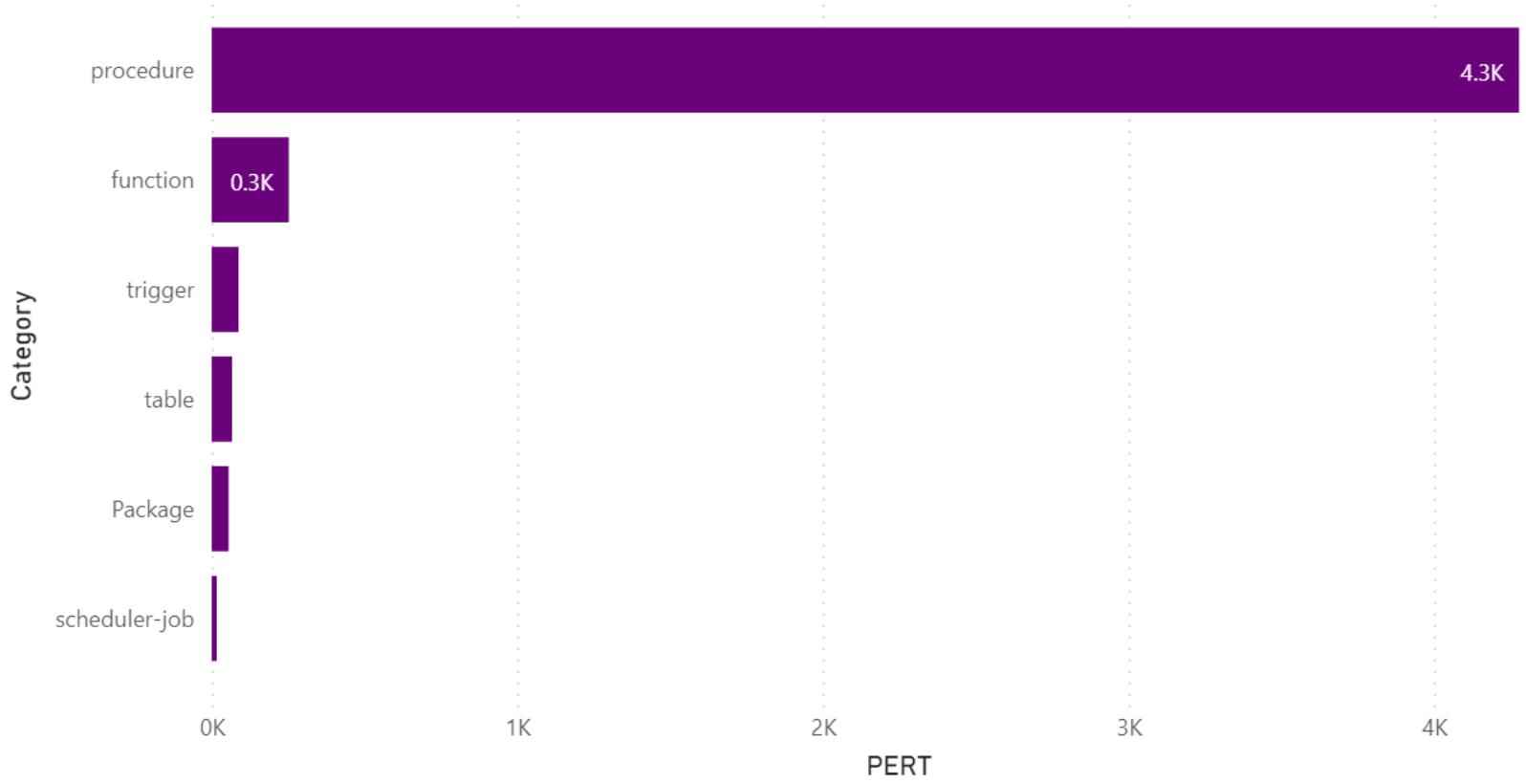
PERT =(O+(4*M)+P)/6

PERT Estimate

4572 Total Hours

Assessment Results

Estimated migration effort



Procedures

4278 Total Hours

Functions

252 Total Hours

Triggers

87 Total Hours

Tables

66 Total Hours

Packages

55 Total Hours

POC Details

We have selected the follow set of manual action groups and objects for the PoC.

Issues Group	Object Name	Object Type	Lines of Code	Simple Actions	Medium Actions	Complex Actions	TC Act
Transaction control is only possible from the top level or nested call invocations without any other intervening comman	H33_UPDATE_RC_PATIENT	procedure	400	10	0	3	10
Transaction control is only possible from the top level or nested call invocations without any other intervening comman	H33_DELETE_PATIENT	procedure	345	5	0	4	2
A transaction cannot be ended inside a block with exception handlers	H33_ADD_PATIENT	function	1520	20	0	0	2
PostgreSQL doesn't support SAVEPOINT, ROLLBACK TO SAVEPOINT inside of routines	H33_DEDUP_PATIENTS_DEBUG	procedure	5608	69	0	2	3
Unable to convert functions	H33_FAC_ORG_SEND	procedure	27	0	0	1	0
PL/SQL pragma is unsupported	H33_GET_LATEST_SEROLOGY_DATE	function	91	0	0	1	0
Dynamic SQL table dependency	H33_START_AUTO_PROCESS	procedure	91	15	0	1	4
Table contains LOB column(s)	H33_HL7_SEND_ERROR	table	0	2	0	0	0
Unable to convert the UPDATE statement with multiple-column subquery in SET clause	H33_FORECAST_VACCINATIONS	procedure	6310	7	0	34	1
Unable to convert statements with dynamic SQL statement	H33_START_AUTO_PROCESS	procedure	91	15	0	1	4
				190			26

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